RESEARCH ARTICLE November 1, 2026

Title of the Article

A Concise Subtitle That Clarifies the Scope

First Author[1] iD 0000-0000-0000-0001 Second Author[1,2] iD 0000-0000-0000-0002
Third Author[2,*] iD 0000-0000-0000-0003

[1] Department of Mathematics, University Name, City, Country
[2] Institute of Applied Sciences, Institution Name, City, Country
[*] Corresponding author: Third Author, third.author@example.org

ABSTRACT

Replace this paragraph with a concise summary of the problem, the method and the main result, in about 150 words. State what was done, what was found and why it matters, and avoid citations or abbreviations that are not explained within the abstract itself. This template sets the article in two balanced columns with a full-width title block, numbered equations, theorems, figures, tables and a reference list.

KEYWORDS keyword one; keyword two; keyword three; keyword four

1 INTRODUCTION

Scientific articles begin by motivating a problem and stating, as plainly as possible, what the reader will learn [1, 2]. Replace this text with the motivation, the gap in the literature and the main contribution.[1] The main contributions are:

- a formulation of the problem (Section 3);
- a convergence guarantee (Theorem 4.1);
- an empirical comparison (Table 1).

The paper is organized as follows. Section 2 reviews related work and fixes notation, Section 3 describes the method, Section 4 reports results, and Sections 5 and 6 discuss and conclude.

2 BACKGROUND AND RELATED WORK

Early approaches treated the problem in isolation [3], while later work unified it with a broader family of models [2]. We write $\|\cdot\|$ for the Euclidean norm and $\langle \cdot, \cdot \rangle$ for the inner product on $\mathbb{R}^d$.

Definition 2.1 (Smoothness)**.** A differentiable function $f\colon \mathbb{R}^d \to \mathbb{R}$ is *L-smooth* if $\|\nabla f(x) - \nabla f(y)\| \le L\|x - y\|$ for all $x, y \in \mathbb{R}^d$.

3 METHODS

3.1 Objective

We fit a parametric model $f(\cdot\,; \theta)$ by minimizing the regularized empirical risk

$$J(\theta) = \mathcal{L}(\theta) + \lambda\, \Omega(\theta), \tag{3.1}$$

where the data term and the regularizer are

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \ell\big(y_i, f(x_i; \theta)\big), \tag{3.2}$$

$$\Omega(\theta) = \tfrac{1}{2}\|\theta\|^2. \tag{3.3}$$

Equation (3.1) combines (3.2) and (3.3) through the weight $\lambda > 0$.

3.2 Optimization

Our analysis relies on a standard inequality for smooth functions.

Lemma 3.1 (Descent lemma)**.** *If f is L-smooth (Definition 2.1), then for all $x, y \in \mathbb{R}^d$*

$$f(y) \le f(x) + \langle \nabla f(x), y - x \rangle + \tfrac{L}{2}\|y - x\|^2.$$

Proof. Put $g(t) = f(x + t(y - x))$. Then $g'(t) - g'(0) \le Lt\|y - x\|^2$ by Cauchy–Schwarz and smoothness. Integrating over $[0, 1]$ gives $g(1) \le g(0) + g'(0) + \frac{L}{2}\|y - x\|^2$, which is the claim. ∎

4 RESULTS

The main result concerns plain gradient descent on Eq. (3.1).

Theorem 4.1 (Convergence)**.** *Let J be L-smooth and bounded below by $J^\ast$. Gradient descent with step size $1/L$, that is $\theta_{k+1} = \theta_k - \frac{1}{L}\nabla J(\theta_k)$, satisfies*

$$\min_{0 \le k < K} \|\nabla J(\theta_k)\|^2 \le \frac{2L\,\big(J(\theta_0) - J^\ast\big)}{K}.$$

[1] Footnotes are set at the foot of the column in which they are cited; keep them short.

Table 1. Mean test error over five runs (illustrative values).

Method	Error (%)	Time (s)	Params
Baseline	12.4	1.8	0.9M
Method A	10.9	2.1	1.1M
Proposed	**9.7**	2.4	1.2M

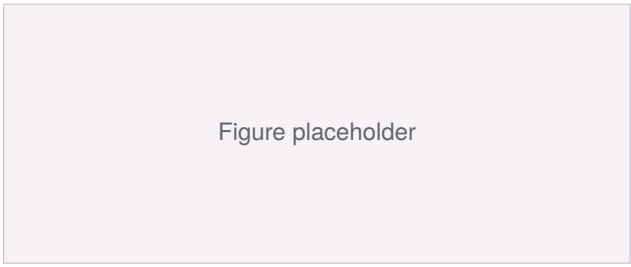

Figure 1. Gradient norm versus iteration (placeholder).

Proof. Apply Lemma 3.1 with $x = \theta_k$ and $y = \theta_{k+1}$:

$$J(\theta_{k+1}) \le J(\theta_k) - \tfrac{1}{2L}\|\nabla J(\theta_k)\|^2.$$

Summing over $k = 0, \dots, K-1$ and telescoping gives $\tfrac{1}{2L}\sum_{k<K}\|\nabla J(\theta_k)\|^2 \le J(\theta_0) - J^*$. The minimum is at most the average, which proves the claim. ∎

Table 1 compares the proposed method with two baselines, and Figure 1 summarizes the behavior over the iterations. The values shown are illustrative; replace them with your own data.

5 DISCUSSION

Interpret the results in light of the question posed in the introduction and compare them with prior work [3]. Theorem 4.1 guarantees convergence to a stationary point only; it says nothing about global optimality. Discuss limitations, threats to validity and the assumptions on which the analysis depends.

6 CONCLUSION

Summarize the contribution in two or three sentences and outline directions for future work.

ACKNOWLEDGMENTS

The authors thank colleagues and reviewers for their helpful comments.

FUNDING

This work was supported by [funding body] under grant number [000000].

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this article.

DATA AVAILABILITY

The data and code supporting the findings of this study are available at `https://example.org/repository`.

REFERENCES

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[2] C. Lee, *Title of a book*, Publisher, City, 2021.

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